

End of flight simulations

The ATSB report [*MH370 – Definition of Underwater Search Areas, December 2015*](#) outlined the previous simulations that the manufacturer had undertaken to assist in determining the aircraft's behaviour at the end of the accident flight.

In April 2016, the ATSB defined a range of additional scenarios for the manufacturer to simulate in their engineering simulator. Reasonable values were selected for the aircraft's speed, fuel, electrical configuration and altitude, along with the turbulence level.

The results of the simulation are presented in Figure 6. The results have all been aligned to the point two minutes after the loss of power from the engines. This is the theorised time at which the 7th arc transmissions would have been sent.

Figure 6: Results from simulated scenarios



Source: ATSB

This figure illustrates the resulting flight paths from the simulations performed by the manufacturer and aligned at a point consistent with when the final BTO transmission may have occurred.

The simulations were completed in the manufacturer's engineering simulator. The engineering simulator uses the same aerodynamic model as a Level D simulator used by the airlines. The simulator is not a full motion simulator but instead is used when a high level of system fidelity is required. The appropriate firmware and software applicable to the accident aircraft can be loaded.

The results of the simulations were that:

- The aircraft was capable of travelling rearwards (from the direction of travel) approximately 21 NM.
- Simulations that experienced a descent rate consistent with the ranges and timing from the BFO analysis generally impacted the water within 15 NM of the arc.
- In some instances, the aircraft remained airborne approximately 20 minutes after the second engine flameout.

- In an electrical configuration where the loss of engine power from one engine resulted in the loss of autopilot (AP), the aircraft descended in both clockwise and anti-clockwise directions.
- In some simulations, the aircraft exhibited phugoid motion² throughout the descent.
- Simulations that exhibited less stable flight resulted in higher descent rates and impact with water closer to the engine flameout location. In some simulations, the aircraft's motion was outside the simulation database. The manufacturer advised that data beyond this time should be treated with caution.
- Some of the simulated scenarios recorded descent rates that equalled or exceeded values derived from the final SATCOM transmission. Similarly, the increase in descent rates across an 8 second period (as per the two final BFO values) equalled or exceeded those derived from the SATCOM transmissions. Some simulated scenarios also recorded descent rates that were outside the aircraft's certified flight envelope.
- The results of the scenarios, combined with the possible errors associated with the BTO values indicate that the previously defined search area width of ± 40 NM is an appropriate width to encompass all uncontrolled descent scenarios from the simulations.

The simulated scenarios do not represent all possible scenarios, nor do they represent the exact response of the accident aircraft. Rather, they provide an indication as to what response the accident aircraft may have exhibited in a particular scenario. As such, the results are treated with caution, and necessary error margins (or safety factors) should be added to the results.

It was not possible to simulate all likely scenario conditions due to the limitations of the simulator. Specifically, flight simulators are unable to accurately model the dynamics of the aircraft's fuel tanks. In the simulator, when the fuel tank is empty, zero fuel is available to all systems fed from the tank. However, in a real aircraft, various aircraft attitudes may result in unusable fuel (usually below engine/APU inlets) becoming available to the fuel inlets for the APU/engines. If this resulted in APU start-up, it would re-energise the AC buses and some hydraulic systems. This could affect the trajectory of the aircraft. Similarly, the left and right engines may also briefly restart, affecting the trajectory.

² A long-period oscillation of pitch axis, perpetually hunting about level attitude and trimmed speed.